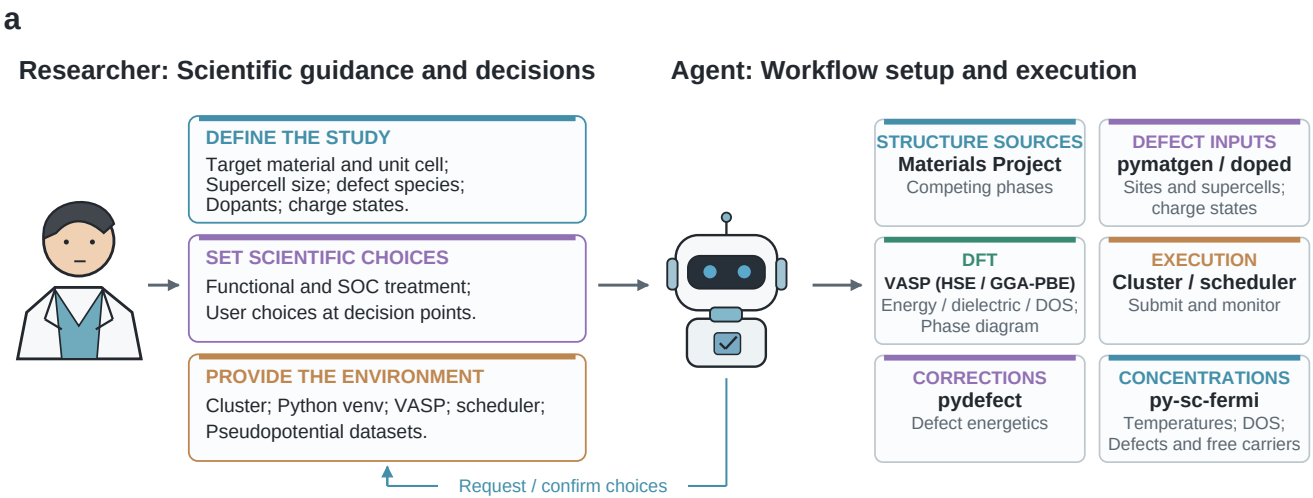

b Defect calculation and analysis workflow

HOST REFERENCES	DEFECT PREPARATION	DEFECT CALCULATIONS	RESULTS AND BACKUP
1 **Unit-cell structure and electronic structure** Relaxation; band gap; DOS	4 **Supercell and defects** Symmetry-inequivalent sites; dopants; charge states	7 **Defect relaxation** Atomic relaxation; fixed lattice Spin polarization; symmetry off	10 **Postprocessing** Charge corrections and formation-energy diagrams
2 **Dielectric constant** Electronic and ionic contributions	5 **Validate and execute** Check inputs; submit and monitor	8 **Static calculation** Final total energies of defect and defect-free supercells	11 **Defect thermodynamics** Temperature; charge neutrality; Defect and carrier concentrations
3 **Phase stability** Formation energies; Elemental chemical potentials	6 **Interstitial screening** GGA-PBE; select distinct minima	9 **Validation** Inputs; outputs; consistency; provenance	12 **Data backup** Preserve hierarchy; verify checksums

◇ **Shared control at every calculation stage**
Validate inputs, check convergence, and resume unfinished relaxations.

○ User decision / confirmation ◇ Validation / recovery checkpoint